CulinaryCut: A Physics-Grounded Cutting Benchmark for Vision-Language-Action Models

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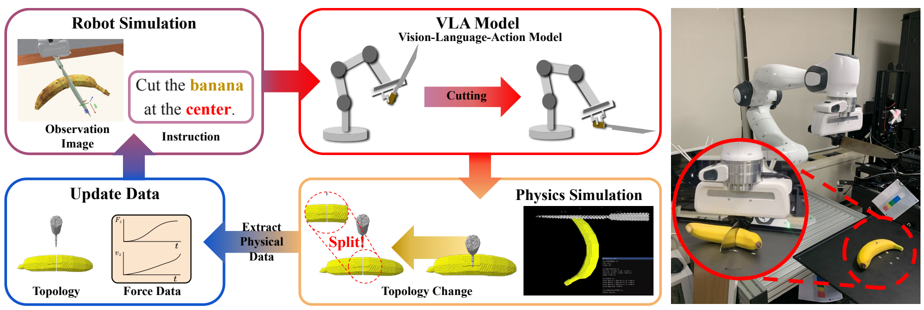

(a) Simulation training phase

(b) Real world deployment

Fig. 1: Physics-grounded pipeline for robotics manipulation. (a) During the simulation stage, we employ a physics simulator to generate force-dependent interactions and incorporate physical signals, including topology changes and contact-force profiles, into the training process. (b) To facilitate robust cutting in real-world environments, we incorporate physical properties into the simulation dataset. *CulinaryCut* provides a standardized benchmark for evaluating these capabilities.

Abstract. Food cutting is a representative non-rigid manipulation task involving deformation, contact forces, and material separation, yet it remains largely unexplored in VLA research. Existing robot manipulation datasets are primarily built around rigid objects and struggle to jointly capture the spatial precision, deformation, topology changes, and force interactions required for cutting. To systematically study these challenges, we introduce CulinaryCut, a benchmark that integrates an MPM-based deformable simulator with a robot environment to generate 325K training instances from 5,000 physically distinct cutting trajectories. Each trajectory is paired with language instructions, multi-view observations, and contact-force profiles, with physical plausibility assessed against force measurements from a real Franka arm. Importantly, these

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forces are embedded in the dataset rather than provided as policy inputs, enabling evaluation of existing VLAs while exposing where physical grounding is needed. Using three representative VLA baselines, we identify two core limitations in non-rigid cutting. First, a geometry gap: VLAs struggle to map ratio- and direction-based instructions onto object geometry, especially when sequential cuts change the object's topology. Second, a physics gap: because policies output motion without a notion of material stiffness or contact resistance, a geometrically accurate path may still fail to sever the object. We show that training VLAs on physics-grounded trajectories, without modifying the policy architecture or adding force inputs, improves cutting success and sim-to-real transfer. Together, CulinaryCut provides a foundation for evaluating spatial reasoning and physical plausibility in deformable manipulation.

Keywords: Vision-Language-Action Model · Deformable Object Manipulation · Physics-Based Simulation

1 Introduction

Vision-Language-Action (VLA) models have achieved strong results in robotic manipulation by integrating visual perception, language understanding, and action generation into a single policy pipeline [4, 5, 7, 21, 29, 34]. Much of this progress, however, has been demonstrated on manipulation tasks involving rigid objects, where success is often determined by spatial alignment, object pose, or reaching a target configuration. These tasks are well suited for evaluating visual grounding and action execution, but they do not fully capture manipulation problems in which task success depends on the physical interaction generated during the motion.

Food cutting is a representative example of such a contact-rich task. To cut a material, a knife must not only follow a plausible geometric path, but also generate sufficient contact forces relative to the object's stiffness, geometry, and internal structure. When an object deforms, separates, or resists the tool, the same nominal motion can lead to different outcomes depending on the force profile it induces, as illustrated in Fig. 1. Thus, a trajectory that appears geometrically correct may still fail to sever the material. Despite being a fundamental everyday skill, food cutting remains largely unexplored in VLA research.

This limitation is closely tied to the data on which current VLAs are trained and evaluated. Standard robot demonstrations typically encode the executed motion, but not the material-dependent contact interactions required for successful cutting. Existing large-scale robot learning datasets focus mainly on rigid-body manipulation [11, 27, 33, 39, 49] and do not capture the forces, deformations, or topological changes that arise during cutting, as summarized in Table 1. Cutting-specific simulators and datasets, such as DiSECt [17], TopoCut [50], SliceIt! [2], and RoboCook [41], model fine-grained material behavior, but they do not jointly provide the physics-grounded contact signals, language instructions, multi-view observations, and continuous robot control trajectories required for VLA training and evaluation.

Recent efforts have begun to incorporate physical cues into robot policies. For example, ForceVLA [56] explores force-conditioned policies by using force information as an additional input. In contrast, our goal is not to design a specialized force-conditioned architecture, but to study whether standard VLA models can learn physics-grounded cutting behaviors when physics is incorporated at the dataset generation stage. This requires a benchmark in which physical interaction, not only geometric motion or visual realism, is faithfully represented.

In this paper, we introduce *CulinaryCut*, a physics-grounded benchmark for food cutting with language-conditioned robot policies. *CulinaryCut* couples an MLS-MPM-based physics simulator [18] with the ManiSkill robot simulator to generate 5,000 cutting trajectories across fifteen food items. Through language augmentation, these trajectories are expanded to 325K training instances, each paired with multi-view observations, continuous control actions, language instructions, and contact-force profiles. To assess physical plausibility, we compare simulated contact-force curves against measurements collected from a real Franka robotic arm.

Using *CulinaryCut*, we evaluate three representative VLA models: RDT-1B [34], Octo [46], and OpenVLA [29]. Our analysis reveals two core limitations. First, models exhibit a geometry gap: they struggle to perceive object extent and ground ratio- and direction-based instructions, especially for small objects and multi-object scenes. Second, models exhibit a physics gap: even when the predicted trajectory is geometrically plausible, it may fail to cut the material because the policy does not account for material stiffness or contact resistance. We show that this physics gap can be reduced without modifying the policy architecture or providing force inputs at inference time, by training on physics-grounded cutting data, improving both cutting success and sim-to-real transfer.

Based on these observations, our contributions are as follows:

1. **Physics-grounded cutting benchmark.** We introduce a benchmark that integrates an MPM-based physics simulator with a robot simulator, providing a large-scale food-cutting dataset and evaluation protocol with force profiles, multi-view observations, continuous actions, and language instructions.
2. **Analysis and data-level mitigation of geometry and physics gaps.** We identify a geometry gap in spatial instruction grounding and a physics gap in material-dependent cutting dynamics, and show that training on physics-grounded trajectories improves cutting success without modifying the policy architecture or providing force inputs at inference time.
3. **Scalable data generation pipeline.** We combine LLM-based language-instruction synthesis with simulation-driven trajectory augmentation to support large-scale dataset construction.
4. **Physical plausibility assessment.** We compare simulated contact-force curves against measurements from a real Franka robotic arm on a representative cutting task.

2 Related Work

Table 1: Comparison of representative robot manipulation and cutting datasets relevant to VLA-based food cutting. Large-scale manipulation datasets may provide language-conditioned or continuous robot trajectories but generally lack cutting support, while cutting-specific simulators model material behavior but do not jointly provide language instructions, multi-view observations, and continuous actions required for VLA training and evaluation.

Dataset	**Physics Sim**	**Robot Sim**	**Data Origin**	**Multi Camera**	**Language Instruction**	**Continuous Action**	**Cutting Support**
DROID [27]	–	–	Real	✓	✓	✓	–
Arnold [14]	–	IsaacSim	Synthetic	✓	–	✓	–
RoboChop [12]	–	–	Real	–	–	–	✓
RoboCook [41]	MPM	–	Hybrid	✓	–	–	✓
Jamdagni et al. [23]	FEM	–	Synthetic	–	–	–	✓
SliceIt! [2]	FEM	Gazebo	Synthetic	–	–	–	✓
DiSECt [17]	FEM	–	Synthetic	✓	–	–	✓
TopoCut [50]	MPM	–	Synthetic	✓	–	–	✓
Ours	**MPM**	**ManiSkill**	**Synthetic**	✓	✓	✓	✓

2.1 VLA Models for Robotic Manipulation.

VLA (Vision-Language-Action) models have rapidly evolved toward unifying perception, language, and action within a single policy backbone. Early approaches primarily focused on behavior imitation [48], and later expanded toward language-conditioned generalist agents [40]. Recent models such as RT-2 [5], OpenVLA [29], RDT-1B [34], Octo [46], π_0 [4], and GR00T N1 [3] have advanced VLA architectures toward large-scale data-driven learning and cross-embodiment generalization. In parallel, reasoning-augmented VLAs incorporate self-talk, VLM-to-VLA transfer, and long-horizon task planning [19,31,45,57,58], and trajectory-centric designs directly map language to motion and control token sequences [8, 25, 53]. Despite this progress, existing VLAs and visuomotor baselines such as Diffusion Policy [10] and ACT [59] have been primarily trained and evaluated on rigid-object manipulation, and their capabilities remain insufficiently validated on contact-rich tasks such as cutting, where complex physical interactions, deformation, and material separation are central. ForceVLA [56] recently explored force-conditioned policies by using force cues as additional inputs; in contrast, we incorporate physics into the dataset generation process, enabling standard VLAs to learn and be evaluated on physics-grounded cutting behaviors without requiring a specialized force-conditioned architecture.

2.2 Benchmarks and Datasets for Manipulation.

This limitation is partly attributable to the lack of suitable training and evaluation data for contact-rich cutting tasks. Large-scale robot learning datasets, such

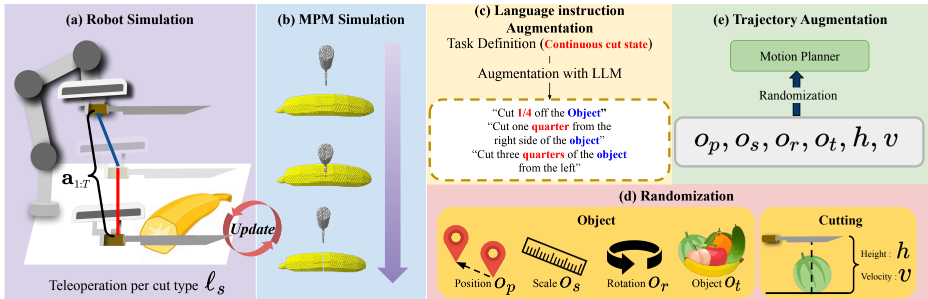

Fig. 2: Overview of CulinaryCut data generation pipeline. (a) Teleoperation generates initial cutting demonstrations for each cut style. (b) Physics simulation updates material deformation and cutting geometry. (c) Language instructions are augmented using LLMs with continuous cut state variations. (d, e) Trajectories are augmented via motion planning with object and cutting randomization.

as Open X-Embodiment [39], DROID [27], BridgeData V2 [49], LIBERO [33], and RoboMIND [54], provide diverse real-robot trajectories but do not include cutting; moreover, collecting large-scale real cutting data is difficult due to safety concerns, material variability, and the need for precise force control. Simulation offers an alternative, yet widely used benchmarks such as ManiSkill [15] do not address cutting, and deformable-manipulation benchmarks such as PlasticineLab [20] and SoftGym [32] do not cover cutting or material separation. Cutting-specific datasets and simulation studies, including DiSECt [17], TopoCut [50], SliceIt! [2], and RoboCook [41], focus on physics simulation and material behavior but do not jointly provide the language instructions, multi-view observations, and continuous control actions required for VLA training.

2.3 Simulation Realism and Physics-Grounded Cutting.

Since our dataset is simulation-based, it inevitably involves a sim-to-real gap. A major component is the visual domain gap [47], for which recent foundation world models offer tools for improving visual realism and diversity. Models such as Cosmos [37], Genie [6], UniSim [55], and GenAug [9] improve visual realism by generating photorealistic or diverse scenes from structured inputs.

However, visual realism alone cannot resolve the physics gap. In cutting, contact forces, deformation, and topology changes are central to success, so realistic imagery is insufficient unless the simulation also encodes physically plausible contact-force signals and material state transitions. Cutting simulation commonly relies on physics-based methods such as the Material Point Method (MPM) and the Finite Element Method (FEM). MPM naturally handles large deformation and topology changes by coupling Lagrangian particles with an Eulerian grid [43, 44], with recent extensions supporting fracture modeling through continuum damage mechanics [51, 52]. FEM-based approaches have been explored for differentiable cutting and robotic slicing [2, 17, 23], and related work in surgical robotics addresses deformable tissue cutting with similar challenges in contact, topology change, and force-sensitive interaction [13, 16].

Among these approaches, we adopt MLS-MPM [18] for its stable handling of large deformation and topology change while preserving rotational and shear responses inherited from APIC [24], and couple it with a robot simulator to build *CulinaryCut*.

3 CulinaryCut Benchmark

3.1 Overview

CulinaryCut is not a new model architecture, but a benchmark designed to diagnose the structural limitations of VLA models in non-rigid cutting tasks. In this paper, we target two such limitations. The ***geometry gap*** denotes the difficulty of grounding spatial, ratio-based, and directional instructions, such as "cut one-quarter from the right," in an object's actual geometry, especially when each cut changes the object's topology and requires subsequent cut positions to be recomputed. The ***physics gap*** captures the challenge of reasoning about the material-dependent physical interaction required to sever a deformable object. Many existing VLAs are primarily trained and evaluated as visual or geometric trajectory-following systems, without explicitly modeling force feedback, contact resistance, or material failure; as a result, a geometrically accurate cutting path alone does not guarantee a successful cut.

To diagnose these limitations, CulinaryCut encodes physics at the dataset level. Each trajectory includes contact-force signals produced by our MLS-MPM simulator; these signals are embedded in the dataset rather than provided as inputs to the VLA policy. This design isolates the geometry gap through the task structure described in Sec. 3.2 and operationalizes the physics gap through force-labeled cutting trajectories. CulinaryCut combines three modules: an MLS-MPM-based physics simulator, the ManiSkill robot simulation environment, and an LLM-based language-instruction generation pipeline. Together, they produce 5,000 cutting trajectories, corresponding to 325K instances after language augmentation, with contact-force signals, multi-view images, and language instructions. This enables quantitative evaluation of spatial reasoning while providing physics-grounded signals for analyzing model behavior under material-dependent cutting dynamics. Fig. 2 shows the full benchmark generation pipeline.

3.2 Task Design

The tasks in CulinaryCut are designed with two main objectives: evaluating ratio-based spatial grounding and assessing sequential planning under topology changes.

Table 2: Material properties used in CulinaryCut. Values are collected from the food-engineering literature and used to parameterize deformable objects in the MLS-MPM simulator.

Object	Density (ρ)	Young's Modulus (E)	Poisson's Ratio (ν)	Reference (Source)
Apple	850 kg/m^3	3.0×10^6 Pa	0.33	Kheiralipour et al. [28]
Cucumber	980 kg/m^3	1.0×10^6 Pa	0.35	Mattos et al. [35]
Melon	1100 kg/m^3	1.2×10^6 Pa	0.45	Nourain et al. [36]
Banana	960 kg/m^3	7.0×10^3 Pa	0.35	Jahanbakhshi et al. [22]
Orange	1020 kg/m^3	5.84×10^5 Pa	0.24	Kurniawati et al. [30]
Peach	968 kg/m^3	8.9×10^5 Pa	0.29	Kabaş & Vladut [26]
Strawberry	1010 kg/m^3	2.0×10^5 Pa	0.40	An et al. [1]

Task for spatial grounding. This task evaluates ratio-based spatial grounding, requiring the agent to translate language-specified proportions into precise cutting positions on the object. In this setting, the language instruction specifies an exact spatial location on the object. Given commands such as "cut the object in half" or "cut one-quarter from the right," the agent must perceive the overall size of the object and convert ratio expressions (1/2, 1/4, 1/6, etc.) into an actual cutting position. Because the same ratio yields different cut locations depending on whether the instruction says "from the left" or "from the right," the ability to correctly ground directional references is also tested. Unless otherwise specified, "left" and "right" are defined with respect to the camera's viewpoint facing the workspace. This task directly measures whether a VLA can perceive object geometry and map ratios onto spatial coordinates.

Task for sequential planning. This task evaluates sequential planning under topology changes, requiring the agent to maintain spatial accuracy over multiple consecutive cuts while adapting to the evolving object geometry. In this setting, the agent must divide an object into a specified number of equal pieces (e.g., "divide the cucumber into four pieces"). Multiple sequential cuts are required, so spatial accuracy must be maintained across every cut. After the first cut changes the object's topology, the remaining cut positions must be recomputed, simultaneously demanding long-horizon planning and adaptation to geometric state changes. Success is measured by the number and uniformity of the resulting pieces.

3.3 Physics-Grounded Simulation

Our simulator is built on MLS-MPM [18], which naturally handles large deformations and topology changes inherent to cutting. Each particle carries position, velocity, deformation gradient, and a damage scalar; corotated elasticity [42] is combined with J2 plasticity and a scalar damage model to represent per-material stiffness and cutting resistance. The fifteen food items are assigned material properties drawn from the food-engineering literature [1, 22, 26, 28, 30, 35, 36]; per-material density, Young's modulus, Poisson's ratio, and their sources are summarized in Table 2. Simulation parameters are initialized from these literature values and may be adjusted for numerical stability.

Crucially, this formulation makes cutting outcomes depend on the contact forces a trajectory induces, not merely on the path it follows. We estimate these forces from the momentum exchange between tool and material: the knife and board are represented as signed distance fields (SDFs), and the velocity change imposed on MPM particles at each grid node is converted into a reaction impulse, yielding the contact force. Because each trajectory thus carries a physics-grounded force-over-time signal, the dataset exposes the physics gap as a measurable property: a geometrically correct path that violates the material's force requirements fails to sever the object, as we show quantitatively in Sec. 4. These force signals also inform the velocity guardrail (Sec. 3.5) and are validated against real Franka measurements. Full equations and setup are in the Supplementary.

3.4 Data Generation

Trajectory generation. Demonstrations for each task are generated by an AABB-based motion planner that dynamically computes the target cutting position from the language instruction. The planner first estimates the spatial extent of the object from its bounding box, then determines the cut location according to the specified ratio and direction. The moment the tool first contacts the object surface is recorded as the contact phase; from that point onward, the physics simulator updates forces, deformation, and topology changes in real time. This process maintains physical consistency between visual observations, actions, and the resulting contact forces, providing physics-grounded supervision for imitation learning.

Dataset statistics. A total of 5,000 continuous trajectories are collected across fifteen food items (banana, cucumber, apple, peach, melon, orange, strawberry, pear, kiwi, tomato, lemon, grape, shine muscat, plum, cherry). The dataset captures object-level diversity through geometric and material variations, and visual scene diversity through background variations. Each trajectory is paired with at least five language-instruction variants, yielding a dataset of 325K total instances.

Language instructions. A template-based language-generation engine synthesizes diverse textual commands for the same manipulation. Each template contains slots for object, position, ratio, and direction, which are randomly filled from a synonym pool (e.g., "50%," "half," "two quarters") to increase linguistic diversity. LLM-generated paraphrases [38] further broaden the range of expressions while preserving semantic content. Because the initial state is not given in text, the agent must infer the current situation directly from visual observation. The full list of templates and lexical variations is provided in the Supplementary Material.

Evaluation protocol. We evaluate VLAs along two complementary dimensions. First, we assess **geometric generalization** along three axes: (1) *unseen random seeds*: object positions and scales not seen during training; (2) *multi-object scenes*: multiple objects coexisting in the same workspace; (3) *unseen language instructions*: commands reserved exclusively for testing. Cross-domain evaluation (e.g., training on apple and evaluating on banana) further measures cross-category transfer. Second, we assess **physical severing** under MPM-coupled dynamics, measuring whether a trajectory induces the material-dependent force required to actually sever the object rather than merely following a geometrically correct path (Sec. 4).

3.5 Baselines and Physics-grounded Comparison

VLA baselines. We evaluate three representative VLA models: RDT 1B [34], Octo [46], and OpenVLA [29]. These models follow the standard VLA paradigm of predicting robot actions from language instructions and visual or multimodal observations, while differing in architecture and training strategy. RDT-1B models action trajectories with a diffusion-based transformer, Octo provides an open generalist policy architecture for cross-embodiment robot learning, and OpenVLA augments a vision-language backbone with an action prediction head. We use these models as representative baselines to test whether current VLA pipelines can handle the geometric and physical requirements of non-rigid cutting. All baselines are fine-tuned on the CulinaryCut training split before evaluation, and none receives force, contact, or damage-state information as policy input.

Physics-grounded data. Our goal is to test whether physics-grounded demonstrations improve cutting behavior without changing the VLA architecture or providing force observations at inference time. To this end, we compare policies trained on kinematic trajectory-only demonstrations against policies trained on our MPM-grounded demonstrations, whose actions are generated under deformable material dynamics and annotated with contact-force signals. The force signal is used to construct and analyze the dataset, but it is not provided as an input to the policy. This comparison isolates whether the physical consistency of the training data, rather than a force-conditioned policy architecture, improves actual cutting success. Quantitative results are reported in Sec. 4.

Velocity guardrail. For fair and safe task-level evaluation, we apply the same velocity guardrail to all baselines at execution time. The guardrail clips commands that would exceed the predefined safe operating range of the Franka arm, but it does not provide additional observations to the policy or re-plan the trajectory. Thus, the policy itself remains force-unaware; the guardrail acts only as an external safety filter applied uniformly across models. Implementation details are provided in the Supplementary Material.

4 Experiments

4.1 Experimental Setup.

Experiment detail All cutting experiments are conducted using publicly available pretrained VLA models that are fine-tuned on the CulinaryCut dataset in a simulated tabletop environment with randomized object placements to evaluate each model's robustness and spatial generalization, and identify failure modes that motivate this benchmark. Before each episode, the target object (e.g., banana, cucumber, apple, melon) is randomly initialized within the workspace boundary on the table. The robot executes the instructed cutting motion according to either a language-based or numerical command. For each object, each task is evaluated over 20 randomized trials (excluding the training seed), where the object's initial pose and position are varied to determine success.

Success criteria A cutting attempt is considered successful if all of the following conditions are satisfied: (1) The knife tip reaches the table surface with a positional error less than 0.05 m. (2) The blade accurately intersects the predefined cutting region of the object. (3) The blade–object contact angle is within $90^\circ \pm 10^\circ$. (4) The cutting location lies within one-tenth of the object size from the designated target position.

Real-world force validation To verify the physical fidelity of our cutting simulation, we compare the end-effector (EEF) force profile from a real Franka Emika Panda robot cutting a banana with the corresponding MPM simulation output.

Real Robot Setup. A Franka Emika Panda arm equipped with a kitchen knife executes a single downward cut on a banana at constant speed. The wrist-mounted F/T sensor records the vertical reaction force at 60 Hz throughout the cutting motion. The sensor reading includes a pre-contact baseline of $\sim$25 N, which we attribute primarily to the static load of the end-effector assembly and sensor offset; this baseline is present before the knife contacts the material and remains approximately constant throughout the non-cutting phases.

MPM Simulation. We replicate the same cutting scenario using our MLS-MPM framework (grid resolution 120^3, $\Delta t{=}2{\times}10^{-5}$ s, 6 substeps). The knife collider moves downward at the prescribed cutting speed, and the contact force is computed from the accumulated impulse on material particles per output interval. Because the simulation models only the knife-material interaction, the output represents pure cutting force with no static offset from a robot arm.

Figure 3 shows the comparison. Both profiles exhibit a consistent pattern: an initial ramp-up as the knife penetrates the material, a peak force region during full-depth cutting, and a decline as the knife exits. The real robot produces a peak cutting force of $\sim$12.5 N above baseline, while the simulation yields $\sim$12.0 N within 5% relative error.

To compare temporal shapes, we subtract the pre-contact baseline from each profile and normalize by its peak. The resulting overlay (Figure 3c) shows close

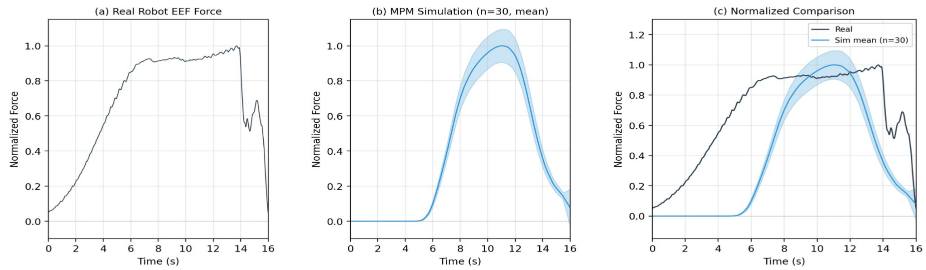

Fig. 3: Real-world force validation. (a) End-effector force measured by the Franka F/T sensor during banana cutting. (b) Contact force from the MPM simulation under the same cutting scenario. (c) Normalized overlay after baseline subtraction and peak normalization, showing close temporal agreement.

Table 3: End-effector force comparison between real robot and MPM simulation.

Metric	Real Robot	MPM Simulation
Pre-contact baseline	∼25 N	0 N (no arm)
Peak cutting force (above baseline)	∼12.5 N	∼12 N

agreement during ramp-up and peak regions, with minor deviations in the retraction phase where the real robot exhibits a broader plateau before force release. These differences are attributable to factors not captured in the current simulation, such as material adhesion, fiber tearing dynamics, and compliance in the real robot's control loop.

4.2 Experiment Results

We summarize overall performance in Table 4 and analyze general, multi-object, transfer, and ratio-based settings with Figures 4–6. Across all tasks, RDT-1B achieves the highest accuracy, while Octo shows the largest degradation under ratio-based instructions, indicating limited numeric grounding. Additional ablation studies are provided in the Supplementary Material.

Performance under object variation. We first report general-task success rate in Table 4. Although average performance appears competitive, Figure 4 reveals notable variance across object categories, with smaller items exhibiting larger error rates. This suggests that current VLAs remain sensitive to object scale even when the instruction and cutting style are unchanged.

Performance under multi-object Scene. To evaluate whether the model can correctly identify and cut the instructed object when multiple objects are present, we conduct a multi-object scene experiment as illustrated in Figure 5.

Table 4: Comparison of average performance of VLA models on cutting tasks (%).

Model	Object Variation	Multi Object	Transfer Object	Continuous Cut	Split Cut
RDT-1B	68.57	31.42	42.85	51.25	0
Octo	42.28	17.00	34.00	6.75	0
OpenVLA	50.57	27.85	26.00	26.87	0

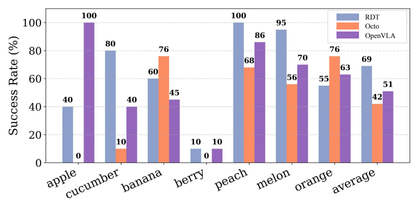

Fig. 4: Object Variation Results. The bar chart shows the inference performance of each model when trained on individual object–instruction pairs.

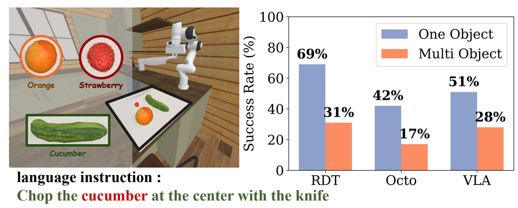

Fig. 5: Multi-Object Results. Comparison of RDT, Octo, and OpenVLA in single- and multi-object scenes. The presence of additional objects significantly impairs target identification and reduces cutting success rates.

Each scene contains multiple fruits placed randomly on the table, and the model receives an instruction referring to one specific target (e.g., "cut the banana at the center"). The results, summarized in Figure 5, show a consistent performance drop across all models when transitioning from single-object (Original) to multi-object scenarios. Specifically, the success rate of RDT decreases from 68.57% to 31.42%, Octo from 42.28% to 17.00%, and OpenVLA from 50.57% to 27.85%. This indicates that while the models possess moderate visual grounding ability, their reasoning and spatial grounding for precise target selection remain limited under complex, cluttered multi-object conditions.

Transfer-object performance. We evaluate the baseline models' ability to generalize to unseen objects when trained on a single-object dataset and tested on different, untrained categories. As shown in Table 4 and Figure 7, performance drops moderately on unseen objects: RDT-1B from 68.57% to 42.85%, and OpenVLA from 50.57% to 26.00%, indicating that the dataset promotes transferable cutting behavior across object categories.

Numeric ratio grounding performance in continuous-cut. As shown in Table 4, our continuous-ratio benchmark reveals a systematic weakness of current VLA-style policies in grounding percentage-based instructions to spatial geometry. Performance generally decreases across most models when given continuous-cut commands (e.g., "cut object 0.40 from right"), compared to the standard "General Task" setting. Even the large diffusion-based baseline (**RDT-1B**) be-

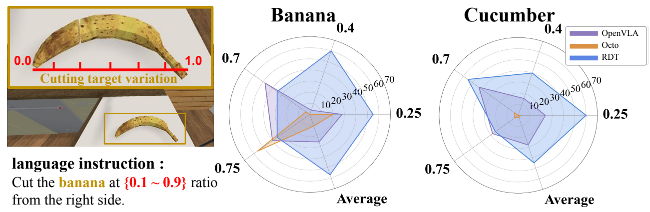

Fig. 6: Evaluation of Continuous Cut. Success rates (%) of RDT, Octo, and OpenVLA under continuous ratio-based cutting instructions. The radar charts show model performance across cut ratios (0.1–0.9) for banana and cucumber.

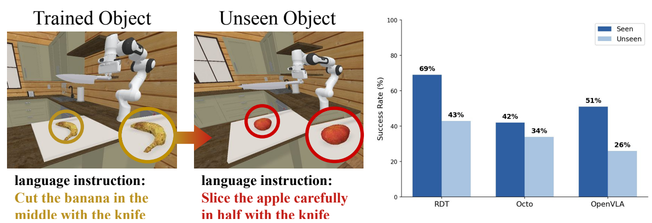

Fig. 7: Generalization to Unseen Objects. Evaluation of model generalization to unseen objects and novel language instructions. The bar chart presents success rates of RDT, Octo, and OpenVLA, showing a clear performance drop on unseen objects.

haves inconsistently across nearby ratio instructions. For instance, on *banana*, the success rate is 60% for "cut object 0.40 from right," but drops to 35% for "cut object 0.70 from right" and "cut object 0.75 from right." On *cucumber*, performance decreases from 60% at "cut object 0.25 from right" to 25% at "cut object 0.75 from right." **OpenVLA** and **Octo** likewise exhibit sharp fluctuations and often collapse at mid-range ratios, as shown in Figure 6, where on *banana* with "cut object 0.40 from right," OpenVLA achieves 5% and Octo 0% success. These results indicate that to correctly execute instructions such as "cut object 0.40 from right," the model must accurately map linguistic numeric ratios to object-centric coordinates and reliably estimate the object's size and spatial proportions.

Table 5: Distribution of successfully placed cuts in Split Cut evaluation with RDT-1B. Each cell shows the percentage of episodes in which exactly k cuts were placed within the $\pm 10\%$ tolerance.

Object	Split Cut	1	2	3	4
Banana	1/3	20%	0%	–	–
	1/4	10%	5%	0%	–
	1/5	5%	0%	0%	0%
Cucumber	1/3	30%	0%	–	–
	1/4	10%	0%	0%	–
	1/5	0%	0%	0%	0%

Sequential division performance in split Cut. As shown in the main table, all models achieve near-zero full success rates under the Split Cut setting. Although some models occasionally place a single cut within the $\pm 10\%$ tolerance, they fail to consistently achieve multiple correct cuts within a single episode, as detailed in Table 5. For instance, under the 1/4 and 1/5 configurations, the probability of correctly placing more

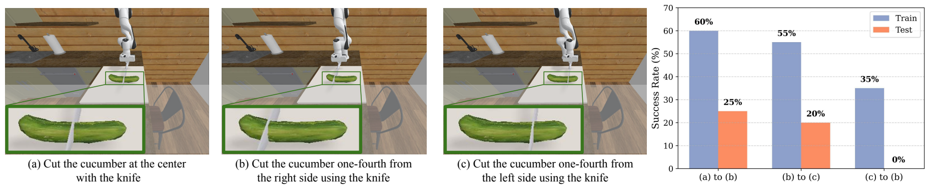

Fig. 8: Transfer Performance Results. (a), (b), and (c) illustrate ratio-based cutting tasks, while the bar chart on the right compares model performance between training and testing conditions, highlighting a clear performance drop on unseen ratio configurations.

than one cut drops close to zero across objects. These results indicate that Split Cut requires more than local ratio grounding; it demands sequential geometric planning across multiple target positions. To divide an object into equal proportions (e.g., $1/4$ or $1/5$), the policy must reason step-by-step about where each subsequent cut should be placed relative to previous cuts and the remaining geometry.

4.3 Ablation Study

Directional and Ratio-based Generalization. We further evaluate the RDT model, which achieved the highest overall performance, to analyze its ability to generalize across directional flips and cutting ratios. As summarized in Figure 8, the model shows a significant degradation when trained on one configuration and evaluated on its directional or proportional counterpart. When the model is trained on 0.25-ratio cuts from the right side and then evaluated on the mirrored left-side instruction (e.g., 'cut the banana at a 0.25 ratio from the left'), its success rate drops from 55% to 20%. Similarly, in the reverse direction ($0.75 \rightarrow 0.25$), the success rate decreases from 35% to 0%, and when transferring from the 0.5-ratio to 0.25-ratio instruction, performance drops from 60% to 25%. Even the strongest diffusion-based policy fails to generalize symmetrically across directional and ratio-based instructions. To bridge this gap, we include a *continuous-cut* split within CulinaryCut that provides dense ratio supervision (0.1–0.9) with explicit left/right labels, enabling continuous grounding of cutting geometry and fostering more robust spatial and linguistic generalization.

5 Conclusion

We presented CulinaryCut, a physics-grounded benchmark for food cutting that integrates an MLS-MPM simulator with a robot simulator to provide 5,000 cutting trajectories (325K instances with language augmentation) spanning seven food items, paired with contact-force profiles and language instructions. Evaluating three representative VLA models reveals two core limitations. The *geometry gap*: models struggle to ground ratio- and direction-based instructions in object geometry, with accuracy degrading sharply on small objects and multi-object scenes. The *physics gap*: because policies output motion without a notion of material stiffness or contact resistance, a geometrically accurate path may still fail to sever the object. We further show that this physics gap can be narrowed not by modifying the policy architecture or adding force inputs, but by training on physics-grounded data, which improves both cutting success and sim-to-real transfer. CulinaryCut thus provides a foundation for evaluating, and beginning to close, the spatial and physical gaps that separate current VLAs from reliable deformable manipulation.

Limitations and Future Directions. CulinaryCut currently covers seven fruit items; extending it to a broader range of food categories (e.g., bread, meat, cheese) and non-food deformable objects would strengthen generality. The MLS-MPM simulator, while validated against real-robot force measurements, does not model viscoelastic relaxation or moisture-dependent material changes, and our force validation currently uses a single representative item; broader multi-material validation remains future work. Our evaluation also focuses on open-loop trajectory execution, leaving closed-loop force-feedback control and adaptive re-planning outside the current scope. Finally, while physics-grounded data addresses the physical realism of the contact dynamics, a visual sim-to-real gap remains. A natural next step is to combine physics-grounded simulation with visually enhanced world models such as Cosmos [37], jointly bridging the visual and physics gaps to generate data that is both photorealistic and physically consistent for more reliable transfer to real-world manipulation.

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